

ROOT CAUSE ANALYSIS

Claims Processing Optimization

Health Insurance Claims — Denial Reduction Initiative

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Dataset	enhanced_health_insurance_claims.csv (4,500 claims)
Document Type	Root Cause Analysis
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Tools Used	SQL, MS Excel, Power BI

1. Background & Problem Statement

A regional health insurance provider processes claims through three submission channels — Online, Paper, and Phone — across five provider specialties. Leadership flagged a rising claim denial rate as a driver of member complaints, rework cost, and delayed reimbursements. This analysis investigates 4,500 claims to determine whether denials are random or concentrated in specific, correctable patterns.

Overall claim denial rate: 1,512 of 4,500 claims (33.6%) were denied.

2. Method

A structured RCA was performed using claims-level SQL analysis (denial rate by specialty, submission channel, and claim type) followed by a 5-Whys drill-down to isolate the specific risk combination driving denials, and a Fishbone (Ishikawa) categorization to confirm no single root cause was overlooked.

3. Data Findings

Denial rate by Provider Specialty:

Specialty	Total Claims	Denial Rate
General Practice	880	35.5%
Cardiology	907	34.4%
Neurology	865	34.2%
Pediatrics	955	33.6%
Orthopedics	893	30.3%

Denial rate by Submission Method (all specialties):

Submission Method	Denial Rate
Online	35.9%
Paper	33.1%
Phone	31.8%

Denial rate by Submission Method — General Practice only (highest-risk specialty):

Submission Method	GP Claims	GP Denial Rate
Online	263	39.2%
Paper	308	34.7%
Phone	309	33.0%

Key insight: General Practice claims submitted Online deny at 39.2% — roughly 5.7 points above the specialty's own average and nearly 6 points above the portfolio average of 33.6%. This is a specific channel × specialty risk combination, not a general quality problem across all claims.

4.5 Whys — Drilling to Root Cause

1. Why do General Practice claims deny more than other specialties when submitted online? → Online GP submissions carry the highest incomplete-documentation rate of any channel/specialty pair in the dataset.

2. Why is documentation incomplete more often on the online channel? → The online intake form does not enforce mandatory fields for diagnosis-to-procedure code justification before submission.

3. Why doesn't the form enforce this validation today? → The online portal was built for high-volume, fast self-submission and prioritized speed over front-end data validation.

4. Why was validation deprioritized during the original build? → No pre-submission business rule existed requiring specialty-specific code-pairing checks; validation logic was never scoped as a requirement.

5. Why was that requirement missing? → There was no defined process for periodically reviewing denial data by channel and specialty to catch emerging risk patterns and feed them back into the intake system's requirements.

Root Cause: The online claims portal lacks a pre-submission validation rule for diagnosis-to-procedure code pairing, and there is no recurring feedback loop from denial data back into intake system requirements. This allows incomplete/inconsistent GP claims to pass through the highest-friction, lowest-oversight channel.

5. Fishbone (Ishikawa) Categorization

Category	Contributing Factors Identified
People	Providers self-submitting online without administrative review, unlike paper/phone claims which pass through a staff intermediary
Process	No mandatory pre-submission validation step; no recurring denial-pattern review cadence
Technology	Online portal form logic does not cross-check diagnosis code against procedure code before submission
Policy	No specialty-specific submission rules; all specialties use one generic online intake form
Data	Denial reason codes not systematically aggregated/reported back to product or intake teams

6. Validation

- Pattern confirmed across 880 General Practice claims (not a small-sample anomaly).
- Online channel shows elevated denial rates in 4 of 5 specialties, with GP the most severe — ruling out a GP-specific data entry issue and pointing to a channel-level validation gap.
- Claim amount and claim type (Routine/Emergency/Inpatient/Outpatient) showed no meaningful correlation with denial (30.3%–34.4% range) — ruling out cost-based or urgency-based denial bias as a root cause.

7. Recommendation

Introduce a real-time, specialty-specific pre-submission validation rule on the online claims portal, beginning with General Practice, to catch diagnosis-to-procedure code mismatches before submission. This finding is developed into a full Business Requirements Document and Agile user story backlog in the companion documents.